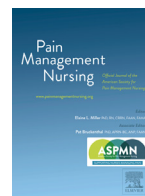




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Review Article

Efficacy of Exercise for Pain Management in Diabetic Peripheral Neuropathy: Network Meta-analysis

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ABSTRACT

Objective: This study aims to compare the efficacy of various exercise modalities through a systematic review and network meta-analysis to determine the optimal exercise regimen for pain in diabetic peripheral neuropathy.

Design: This is a systematic review and network meta-analysis.

Data Sources: Multiple databases were searched, incorporating Web of Science, PubMed, Embase, CINAHL, the Cochrane Library, CNKI, SinoMed, and Wanfang Database.

Methods: To identify eligible randomized controlled trials, a systematic literature search was completed on April 8, 2025. Study quality and potential bias were evaluated using the Cochrane Risk of Bias (version 2). The effectiveness of different interventions was assessed by pairwise and network meta-analysis.

Results: A total of 18 randomized clinical trials were ultimately included in this study, encompassing 1,121 patients. The pairwise meta-analysis revealed that multi-component exercise significantly decreased diabetic peripheral neuropathy pain, and whole-body vibration exercise demonstrated positive analgesic effects in these patients. Network meta-analysis ranked strengthening exercises as the most effective, followed by whole-body vibration.

Conclusions: Our study confirms the superiority of multi-component exercises and recommends strengthening and whole-body vibration exercises as core strategies for evidence-based pain management in diabetic peripheral neuropathy.

Nursing Practice Implications: Our study identifies the core strategies for pain management, aiming to optimize the overall health and well-being of patients with diabetic peripheral neuropathy.

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Key Practice Points

- Our work has clarified the significant analgesic effects of multi-component exercise, which is of guiding significance for alleviating pain in diabetic peripheral neuropathy.
- Furthermore, among these exercises, resistance training and whole-body vibration exercise have shown better analgesic effects, which allows nurses to provide more tar-

geted guidance to diabetic peripheral neuropathy patients for pain relief.

The latest Global Burden of Disease Report indicates that diabetes afflicts approximately 506 million individuals worldwide, with projections indicating that the number of affected individuals will increase to 700 million by 2045 (Zhang et al., 2024). Diabetic peripheral neuropathy, a major chronic complication of diabetes, is associated with neuropathic pain in an estimated 36% of affected patients (Braffett et al., 2024). Neuropathic pain in diabetic peripheral neuropathy is characterized by burning, electric-shock-like, sharp, or dull sensations of variable intensity, imposing sig-

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nificant adverse effects on patients' emotional health and quality of life (Chang & Yang, 2023). Consequently, pain management in diabetic peripheral neuropathy represents a critical clinical issue that warrants ongoing attention to advance evidence-based interventions for this population.

Managing pain associated with diabetic peripheral neuropathy remains a formidable clinical challenge, primarily due to the limited efficacy of current pharmacological interventions and the development of tolerance, which significantly restricts their long-term clinical applicability (Rastogi & Jude, 2021). This predicament underscores the urgent need to explore alternative, non-pharmacological strategies for pain alleviation in patients with diabetic peripheral neuropathy. Among these, exercise has emerged as a promising research focus and is increasingly integrated into clinical practice as a non-invasive approach to mitigate pain in diabetic peripheral neuropathy.

Despite the growing interest in physical exercise, evidence regarding its effectiveness in this domain remains inconclusive. A systematic review and meta-analysis that aggregated two trials revealed no significant between-group differences in pain relief among patients with diabetic peripheral neuropathy (Tatikola et al., 2022). Another scoping review comprehensively summarized various physical activities that can alleviate the pain of diabetic peripheral neuropathy, including aerobic exercise, resistance exercise, whole-body vibration, resistance-aerobic exercise, Tai Chi, and foot-ankle exercises (Pebrianti et al., 2024). While a review suggested that low-intensity resistance training and aerobic exercise offer analgesic benefits (Johnson & Takemoto, 2019), others identified combined resistance-aerobic exercises and whole-body vibration training as effective pain management strategies (Holmes & Hastings, 2021). However, these reviews lack pooled estimates of the overall analgesic effect of exercise and fail to conduct comparisons of different exercise modalities.

To address these knowledge gaps, a comprehensive evidence synthesis integrating diverse exercise interventions and a quantitative network meta-analysis is imperative. Network meta-analysis is a robust statistical method that synthesizes data from multiple studies, enabling both direct and indirect comparisons of two or more interventions (Jansen et al., 2011). This approach facilitates the derivation of comparative efficacy rankings, making it suitable for determining the relative effectiveness of different exercises in managing diabetic peripheral neuropathy-related pain. Therefore, our study will conduct a comparative systematic review to analyze the analgesic efficacy of different exercise modalities in improving pain associated with diabetic peripheral neuropathy, with the core objective of offering evidence-based recommendations for pain management and clinical nursing practice in patients with diabetic peripheral neuropathy.

Methods

This systematic review and network meta-analysis have been prospectively registered via the PROSPERO platform (registration number: CRD420251252585). It was conducted strictly per the methodological specifications outlined in the Cochrane Handbook for Systematic Reviews of Interventions (Higgins, Savović, et al., 2020) and adhered to the reporting requirements of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses statement (Page et al., 2021).

Literature Search Strategy

The literature search as of April 8, 2025, involved multiple databases, incorporating Web of Science, PubMed, Embase, CINAHL, the Cochrane Library, CNKI, SinoMed, and Wanfang Database. Gray

literature was identified from OpenGrey and the reference lists of relevant reviews. The search terms included "diabetic peripheral neuropathy," "exercise," "pain," and "clinical trials." The detailed search strategy, developed using MeSH terms and synonyms, is provided in Supplementary File A.

Inclusion and Exclusion Criteria

The inclusion criteria were as follows: (1) Population: these patients were diagnosed with diabetic peripheral neuropathy; (2) Intervention: any form of exercise modality; (3) Control: other exercise modalities or usual care. Usual care is defined as a control that did not receive any exercise (e.g., treatment as usual, health education); (4) Outcomes: focused primarily on pain relief; (5) Study design: randomized clinical trials. If the quantitative components of the mixed-method study met the inclusion criteria, it was considered. Duplicate publications, non-peer-reviewed articles, papers lacking original data, and studies not published in English or Chinese will be excluded.

Study Selection and Data Extraction

All retrieved records were independently screened for titles and abstracts by two reviewers according to predefined eligibility criteria, and categorized as "included," "uncertain," or "excluded." Full-text assessments were conducted for records marked as "included" or "uncertain." A pre-designed data extraction form was utilized to extract information from eligible studies, including first author, year of publication, country, sample size, intervention details, control group, and pain assessment tools. In accordance with the definitions below, each exercise type in the included studies was labeled to support further data synthesis and statistical analysis. Aerobic exercise is defined as activities enhancing cardiopulmonary function and energy metabolism while foot-ankle exercise is defined as site-specific targeted exercises for the feet and ankles; hand-foot exercise as coordinated movements involving both hands and feet; whole-body vibration as exercises involving whole-body exposure to mechanical vibrations from vibration platforms; balance exercise as activities aimed at improving posture control and balance function; resistance exercise as movements performed against external resistance; and strengthening exercise as exercises designed to enhance muscle strength, endurance, and joint stability. Following data extraction and label, cross-verification was performed by the two reviewers, with discrepancies addressed via a third reviewer.

Quality Appraisal

The Risk of Bias (version 2) from the Cochrane Intervention System Assessment Manual (Higgins, Savović, et al., 2020) was used as the methodological guide for quality assessment in this study. Two reviewers independently evaluated the quality of each included trial based on the randomization process, deviations from intended interventions, missing outcomes, outcome measurement, and selection of reported results. The risk of bias was categorized into five levels: "high risk of bias," "probably high risk of bias," "low risk of bias," "probably low risk of bias," or "no relevant information." Based on the overall assessment of these ratings, the risk of bias was further classified as "low risk," "high risk," or "some concerns."

Data Synthesis and Analysis

To assess the analgesic effects of exercise in diabetic peripheral neuropathy, pairwise and network meta-analyses were conducted.

The standardized mean difference (SMD) and 95% confidence interval (CI) were used as effect size metrics to present the results of the meta-analysis. Statistical heterogeneity was categorized into four levels using the I^2 statistic (Higgins, Thompson, et al., 2003). Subgroup meta-analysis was employed to further examine differences in exercise effects across subgroups and identify sources of heterogeneity. Pairwise meta-analysis and network meta-analysis were conducted under a frequentist framework using the “network” command in STATA 18.0 software (Chaimani et al., 2013). A network diagram was visualized for different exercise modalities in these included studies (Sun et al., 2022). Exercise modalities were represented by nodes in the network diagram, with sample sizes reflected by the size of the nodes. When two interventions are directly comparable, they are represented by solid lines between nodes, whereas unconnected nodes undergo indirect comparison via network relationships. The number of comparative studies is denoted by the thickness of the connecting lines. In our network meta-analysis, the Wald test was employed to assess global inconsistency, and node-splitting analysis was used to examine local inconsistency within loops. A p value exceeding .05 indicates no significant inconsistency, thereby justifying the use of a consistency model (Higgins, Jackson, et al., 2012). The effect estimates of all exercise modalities were summarized in league tables, with ranking identified by the surface under the cumulative ranking curve (SUCRA). The SUCRA values range from 0% to 100%, where higher values denote superior efficacy (Salanti et al., 2011). The Egger’s test and funnel plots were utilized to detect publication bias (Egger et al., 1997), while the trim-and-fill analysis was employed to further evaluate the robustness of the results.

Results

Search Results

A total of 2,719 potentially relevant records were retrieved from electronic databases. Following the exclusion of 513 duplicate records, 2,206 records underwent title and abstract screening, with 2,174 records excluded as ineligible. Ultimately, 32 records were retained for full-text review. Five studies involving non-diabetic peripheral neuropathy populations, seven non-randomized controlled trials, and five duplicate studies were excluded, yielding 18 clinical trials included in our systematic review and network meta-analysis (Cao et al., 2015; Cruvinel-Júnior et al., 2022; Dixit et al., 2014; Jamal et al., 2020; Kessler et al., 2020; Khurshid et al., 2025; Liu & Si, 2017; Luo et al., 2021; Naimat et al., 2022; Reyhanioglu et al., 2024; Serry et al., 2016; Tayebe et al., 2018; Waheed et al., 2021; Wei et al., 2020; Win et al., 2020; J. Yue, 2017; Y. Yue et al., 2023; Zhou, 2019). The selection process is detailed in the Preferred Reporting Items for Systematic Reviews and Meta-Analyses flowchart (Fig. 1).

Trial Characteristics

The details of the included trials are provided in Table 1. These trials were published between 2014 and 2025, with 15 trials conducted in Asia (15/18, 83%), two in the Americas (2/18, 11%), and one in Africa (1/18, 6%). A total of 1,121 patients with diabetic peripheral neuropathy were included. The sample sizes of the included trials ranged from 24 to 120 participants. Eight exercise strategies were identified in intervention groups: aerobic exercise (9/18, 50%), foot-ankle exercise (2/18, 11%), hand-foot exercise (1/18, 6%), whole-body vibration (2/18, 11%), balance exercise (1/18, 6%), resistance-aerobic exercise (1/18, 6%), whole-body vibration-balance exercise (1/18, 6%), and multisystem physical exercise (1/18, 6%). The most commonly used tools for measuring di-

abetic peripheral neuropathic pain were the Visual Analogue Scale (7/18, 39%), followed by the Numeric Pain Rating Scale (3/18, 17%) and the Neuropathy Quality of Life Scale-Pain Score (2/18, 11%).

Risk of Bias Assessment

The details of the bias risk from the included studies are provided in Figure 2. Randomization procedures were clearly described in eight trials (8/18, 44%), whereas concerns were raised regarding 10 studies due to unclear reporting of allocation concealment (10/18, 56%). Most studies showed no deviations from intended interventions (12/18, 67%). Complete outcome data were reported in 12 studies (12/18, 67%). Nine studies demonstrated a low risk of bias in outcome measurement (9/18, 50%), whereas another nine carried a high risk due to inadequate blinding implementation (9/18, 50%). Most studies showed no risk of bias related to selective reporting of results (17/18, 94%). Overall, 12 studies were assessed as high risk, four as having some concerns, and two as low risk of bias. The included studies demonstrated a high overall risk of bias owing to inadequate implementation of blinding and allocation concealment.

Pairwise Meta-analysis

Pairwise meta-analysis was performed across 13 studies (Cao et al., 2015; Cruvinel-Júnior et al., 2022; Dixit et al., 2014; Jamal et al., 2020; Kessler et al., 2020; Khurshid et al., 2025; Liu & Si, 2017; Naimat et al., 2022; Tayebe et al., 2018; Waheed et al., 2021; Win et al., 2020; J. Yue, 2017; Zhou, 2019). The pooled effects of these studies, including the overall effect of exercise on pain relief in diabetic peripheral neuropathy and the effects of each trial, are presented in Supplementary File B. These 13 trials indicated no significant reduction in pain between the two groups (SMD: -0.29 , 95% CI $[-1.10, 0.52]$, I^2 : 96.27%). Results of the subgroup analysis are illustrated in Figure 3. Subgroup analysis revealed that multi-component exercise significantly reduced pain of diabetic peripheral neuropathy (SMD: -0.97 , 95% CI $[-1.38, -0.56]$, I^2 : 0.00%), whereas single-component exercise did not reach statistical significance (SMD: -0.09 , 95% CI $[-1.11, 0.92]$, I^2 : 97.25%). Notably, whole-body vibration exercise exhibited a positive analgesic effect in patients with diabetic peripheral neuropathy (SMD: -1.19 , 95% CI $[-1.82, -0.57]$, I^2 : 0.00%).

Network Meta-analysis

The network meta-analysis was conducted on 11 studies (Cao et al., 2015; Cruvinel-Júnior et al., 2022; Dixit et al., 2014; Jamal et al., 2020; Kessler et al., 2020; Liu & Si, 2017; Naimat et al., 2022; Tayebe et al., 2018; Win et al., 2020; J. Yue, 2017; Zhou, 2019). Six exercise modalities are presented in the network diagram (Fig. 4). The network diagram shows that aerobic exercise had the largest sample size and the most direct comparisons among the included studies. The global inconsistency analysis indicated significant consistency in the evidence network ($\chi^2 = 0.36$, $p = .55$), a consistency model was thus used for data synthesis. No local inconsistency analysis was performed as there were no loops. The network meta-analysis suggested that strengthening exercise (SMD: -5.37 , 95% CI $[-9.52, -1.22]$), whole-body vibration exercise (SMD: -5.16 , 95% CI $[-8.69, -1.62]$), resistance-aerobic exercise (SMD: -5.16 , 95% CI $[-9.21, -1.12]$), aerobic exercise (SMD: -4.55 , 95% CI $[-7.73, -1.38]$), hand-foot exercise (SMD: -4.03 , 95% CI $[-7.53, -0.53]$), were superior to foot-ankle exercise for pain reduction (Supplementary File C). Based on SUCRA results, the most effective exercise for pain reduction was strengthening exercise (74.1%), followed by whole-body vibration exercise (72.1%),

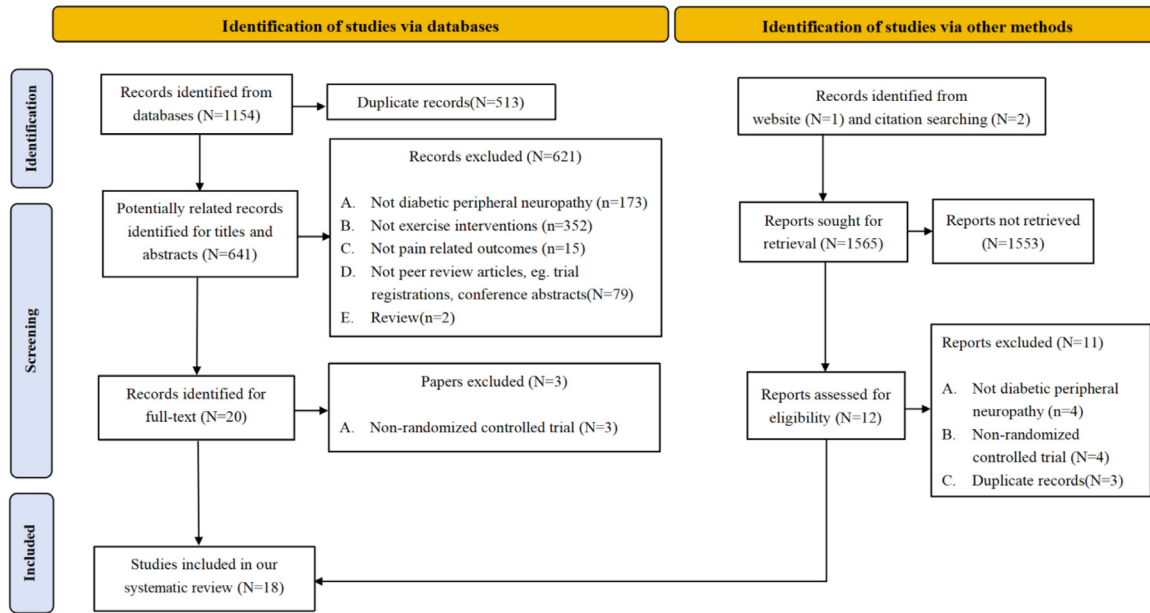


Figure 1. Flow chart of included studies in the systematic review.

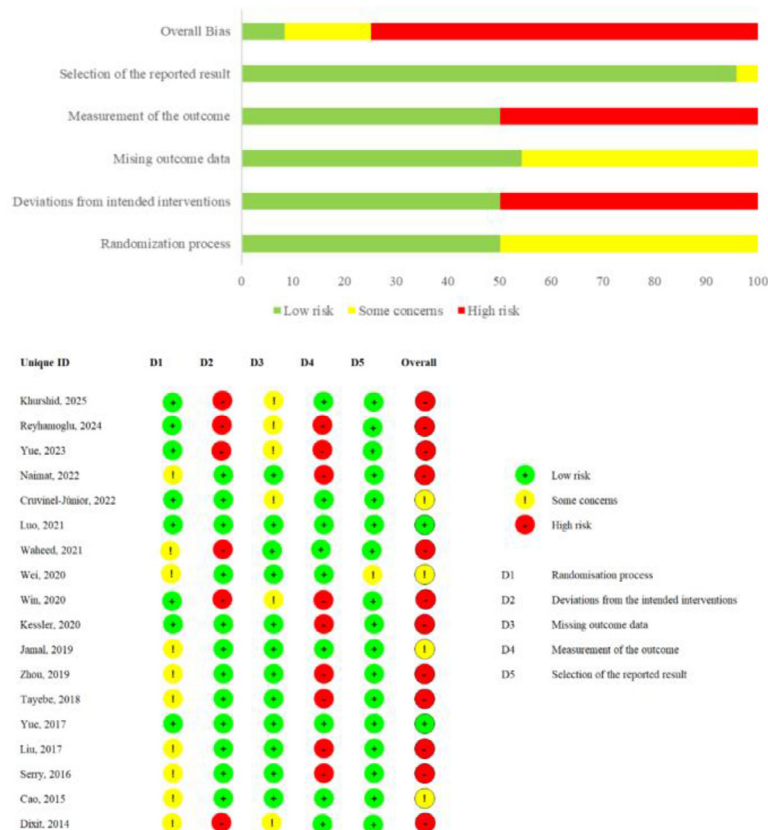


Figure 2. Risk of bias within studies.

resistance-aerobic exercise (68.8%), aerobic exercise (57.0%), hand-foot exercise (42.1%), usual care (35.4%), and foot-ankle exercise (0.6%) (Supplementary File D). No publication bias was detected in the funnel plot ($t = -0.17$, $p = .872$) (Supplementary Files E).

Furthermore, three randomized controlled trials evaluated the efficacy of aerobic exercise in relieving pain associated with di-

abetic peripheral neuropathy, all of which demonstrated statistically significant effects (all, $p < .05$) (Serry et al., 2016; Wei et al., 2020; Y. Yue et al., 2023). A randomized controlled trial indicated that patients in the resistance exercise and foot-ankle exercise groups experienced more pronounced pain reduction compared with the usual care group (all, $p < .05$) (Luo et al., 2021).

Table 1
Characteristics of the Included Studies.

Study ID	Year	Country	Sample Size		Intervention		Controls	Pain Measurements
			IG	CG	Content	Frequency		
Khurshid	2025	Pakistan	25	25	Multisystem physical exercise: reaction time exercise with auditory cues, proprioceptive, muscle strengthening, and balance exercise	8 weeks, 3 sessions per week, 40 minutes per session,	Conventional physical exercise: balance exercise, strength training, stretching, and range of motion exercise	Numeric Pain Rating Scale
Reyhamioglu	2024	Turkey	16	16	Balance exercise	8 weeks, 2 sessions per week, 30 minutes per session	Usual care	Douleur Neuropathique 4 questionnaire Visual Analogue Scale
Yue	2023	China	30	30	Aerobic exercise	12 weeks, 3 sessions per week, 60 minutes per session	Usual care	Visual Analogue Scale
Naimat	2022	Pakistan	40	40	Aerobic exercise	10 weeks, 3 sessions per week, 30 minutes per session	Strengthening exercise	Visual Analogue Scale
Cruvinel-Júnior	2022	Brazil	15	15	Foot-ankle exercise	12 weeks, 3 sessions per week, 20-30 minutes per session	Usual care	Foot Health Status Questionnaire-Foot Pain Score
Luo	2021	China	80	40	Foot-ankle exercise Resistance exercise	12 weeks, 5 sessions per week, 20-30 minutes per session	Usual care	The Frequency of Pain Occurrence
Waheed	2021	India	20	20	Whole-body vibration-balance exercise	3 weeks, 5 sessions per week, 55 minutes per session	Balance exercise	Numeric Pain Rating Scale
Wei	2020	China	50	50	Aerobic exercise	12 months, 3 sessions per week, 20-60 minutes per session	Usual care	The Frequency of Pain Occurrence
Win	2020	Japan	51	53	Hand-foot exercise	8 weeks, 3 sessions per week, 10 minutes per session	Usual care	Behavioral Rating Scale Visual Analogue Scale
Kessler	2020	USA	17	8	Whole-body vibration exercise	4 weeks, 3 sessions per week, 12 minutes per session	Usual care	Visual Analogue Scale
Jamal	2019	India	13	13	Whole-body vibration exercise	6 weeks, 3 sessions per week, 15 minutes per session	Usual care	Numeric Pain Rating Scale
Zhou	2019	China	48	48	Aerobic exercise	3 months, 3-5 sessions per week, 20 minutes per session	Usual care	Visual Analogue Scale
Tayebe	2018	Iran	13	11	Resistance-aerobic exercise	16 weeks, 3 sessions per week, >60 minutes per session	Usual care	Visual Analogue Scale
Yue	2017	China	30	30	Aerobic exercise	8 weeks, 5 sessions per week, 45-60 minutes per session	Usual care	Short-Form Health Survey-Pain Score
Liu	2017	China	40	47	Aerobic exercise	8 weeks, 5-6 sessions per week, 60 minutes per session	Usual care	Neuropathy Quality of Life Scale-Pain Score
Serry	2016	Egypt	20	20	Aerobic exercise	8 weeks, 3 sessions per week, 50 minutes per session	Usual care	Visual Analogue Scale
Cao	2015	China	31	29	Aerobic exercise	Not report	Usual care	Visual Analogue Scale
Dixit	2014	India	40	47	Aerobic exercise	8 weeks, 5-6 sessions per week, 150-360 minutes per week	Usual care	Neuropathy Quality of Life Scale-Pain Score

IG = Intervention Group; CG = Control Group.

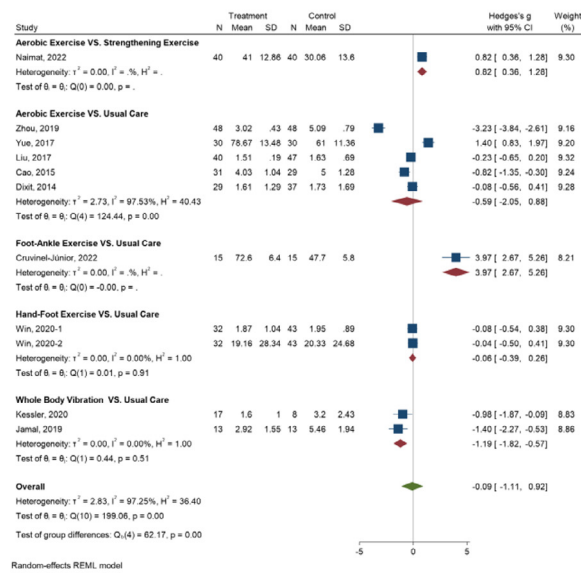
In contrast, another randomized controlled trial found that balance training failed to effectively alleviate pain related to diabetic peripheral neuropathy when compared with standard care ($p = .08$) (Reyhanioglu et al., 2024).

Discussion

This systematic review and network meta-analysis compared the analgesic effects of various exercise modalities in diabetic peripheral neuropathy. Our findings indicated that multi-component exercise programs and whole-body vibration exercise exert significant analgesic effects on pain relief in diabetic peripheral neuropathy patients. Further analysis revealed that, among the eight exercise strategies, strengthening exercise was the most effective for managing pain in diabetic peripheral neuropathy, followed by whole-body vibration exercise. These results suggest that multi-component exercise interventions should be prioritized when designing pain management strategies for diabetic peripheral neuropathy, with strengthening training and whole-body vibration exercise recommended as key components.

Consistent with previous review findings (Tatikola et al., 2022), our study also failed to demonstrate a significant overall effect of exercise on reducing pain in diabetic peripheral neuropathy. The lack of significance may be influenced by complex factors such as exercise components and dosage. Multi-component physical exercise has been recognized as a valuable therapeutic intervention for managing pain of diabetic peripheral neuropathy (Singleton et al., 2015), with combinations of resistance training and sensorimotor training shown to be particularly beneficial for patients with diabetic peripheral neuropathy and other forms of peripheral neuropathy (Streckmann et al., 2022). Resistance-aerobic exercise was identified as effective for relieving pain in diabetic peripheral neuropathy patients, a finding consistent with our study results (Holmes & Hastings, 2021; Pebrianti et al., 2024). Whole-body vibration exercise is considered a viable strategy for diabetic peripheral neuropathy-related pain relief and has been proposed as an alternative to traditional strength training (Holmes & Hastings, 2021). A meta-analysis similar to our study concluded that whole-body vibration exercise improves pain conditions in older adults living with type 2 diabetes (Gomes-Neto et al., 2019). However, prior evidence has shown that prolonged high-frequency

(A) Single-component exercise.



(B) Multi-component exercise.

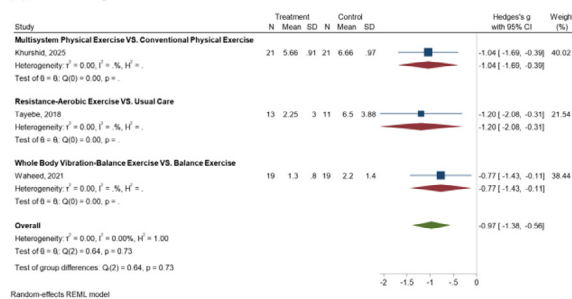


Figure 3. Subgroup meta-analysis of exercise for pain relief.

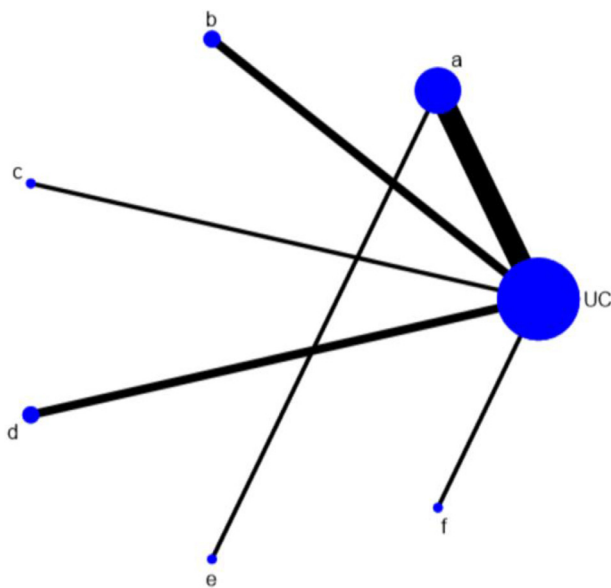


Figure 4. Network diagram of exercise for pain relief. a: aerobic exercise; b: whole-body vibration exercise; c: foot-ankle exercise; d: hand-foot exercise; e: strengthening exercise; f: resistance-aerobic exercise; UC: usual care.

whole-body vibration may carry dangerous side effects, including increased risk of lumbar degenerative changes, lower back pain, muscle fatigue, and female genital disorders (Holmes & Hastings, 2021). Caution is, therefore, warranted in implementing whole-body vibration exercise, necessitating identification of the safest protocols for long-term use. In contrast to our findings, a previous review suggested that low-intensity aerobic exercise alleviates pain in patients with diabetic peripheral neuropathy (Johnson & Takemoto, 2019). This conclusion was based on a single-group self-controlled design with only two supporting trials. The types and quantities of the included studies may be one of the reasons for the different results compared with our study.

It has been hypothesized that exercise alleviates pain by regulating interactions between the immune system and pain pathways, primarily through inhibition of inflammatory responses. Another proposed mechanism involves the release of endogenous analgesic mediators. Preclinical evidence in diabetic animal models has identified abnormal expression of sodium ion channels and multiple molecular dysregulations in sensory neurons, which contribute to pain pathogenesis (Craner et al., 2002; Miyashita et al., 2023). Exercise may activate satellite glial cells in diabetic peripheral neuropathy, which upon activation, release pro-inflammatory mediators that inhibit pain through altering the sensory neuron pathways (Hanani, 2025). Additionally, regular physical activity is thought to stimulate the release of endogenous endorphins and improve systemic circulation, potentially enhancing analgesic effects via peripheral mechanisms (Akhtar, 2024).

To our knowledge, this is the first network meta-analysis to comprehensively compare the analgesic effects of diverse exercise modalities in diabetic peripheral neuropathy. However, this study has several limitations. First, a high risk of bias was identified in the included studies. Most randomized controlled trials included in our study have even demonstrated inadequate blinding and distribution concealment, raising concerns about potential bias. Second, this work focused solely on comparing the efficacy of different exercise types in alleviating diabetic peripheral neuropathy-related pain and did not fully explore exercise frequency, intensity, and time. Future studies should conduct a meta-regression to identify the optimal exercise dose based on frequency, intensity, and time for mitigating diabetic peripheral neuropathy-related pain. Next, two studies from the pairwise meta-analysis were excluded from the network meta-analysis due to insufficient connectivity, leading to insufficient comparisons of multi-component exercise in the analysis. It is recommended that stakeholders' preferences (identified via SUCRA rankings) be integrated (Daly et al., 2020) or that the estimated effects and evidence quality from network meta-analyses be considered (Ostinelli et al., 2022), to facilitate the implementation of exercise interventions for patients with diabetic peripheral neuropathy and extend individual patient benefits to broader populations.

Conclusion

Overall, exercise plays a critical role in pain management for patients with diabetic peripheral neuropathy. Our study underscores the effectiveness of multi-component exercise and encourages the integration of strengthening training and whole-body vibration exercises as core strategies for pain management, aiming to optimize the overall health and well-being of patients with diabetic peripheral neuropathy.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Mei-di Shen: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Rui-tong Gao:** Methodology, Formal analysis. **Si-bing Chen:** Data curation, Conceptualization. **Si-ming Cheng:** Writing – review & editing, Data curation. **Xiang-dong Ding:** Writing – review & editing.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.pmn.2026.04.012](https://doi.org/10.1016/j.pmn.2026.04.012).

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